

Chipathon 2025

Prof James Morbius

Project Proposal
-Updated Version-

PROF MORBIUS

ONE OF THE CHIPS OF ALL TIME



Team Introduction



Team members

- [Leader] James Patrick <@orpheus016:matrix.org>
- [Member] Ibrahim Hanif Mulyana <@ibrhmnhnfm:matrix.org>
- [Member] Goldwin Sonick Wijaya Thaha <@goldwinsonick:matrix.org>

Team background

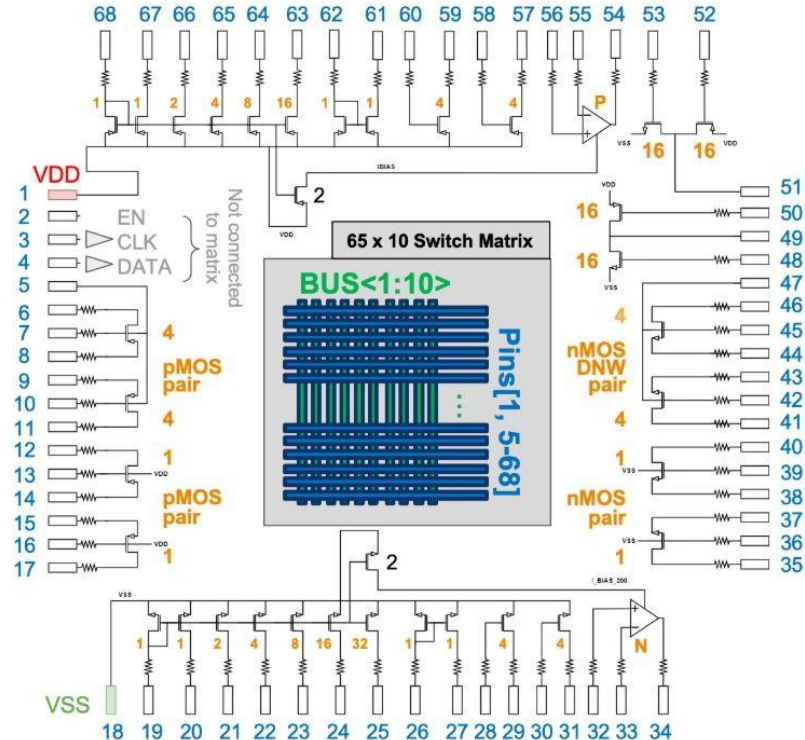
- Experience
 - We are undergraduate student that just finished 3rd year in Electrical Engineering from Bandung Institute of Technology²
 - We have experience designing a chip layout using IIC-OSIC-TOOLS (Xschem, Magic, etc) with SKY130 PDK
 - We have experience designing an RTL using Verilog/VHDL and layout generation using OpenLane2

Project Background and Goals

The MOSbius as a switch matrix can be used to do experiments on analog and RF designs. With wireless communications becoming increasingly used in wider region and scale, we ought to test the reliability of these communication systems, or rather, the reliability of techniques used to ensure our communication to be working. (One of them being spread spectrum techniques, FHSS and DSSS)

By doing this project, our goal is to:

- Design a (Frequency Hopping Spread Spectrum) FHSS Educational Learning Kit with Custom Analog Chip and Comprehensive Learning Guide.
- Design a custom IC with core analog blocks of a frequency synthesizer with system integration to off the chip components (COTS and MOSbius)



Project Information

- **Design**

- On-Chip (Custom IC), we will design 1) **PN Sequence Generator** to control the frequency hopping pattern, 2) **Mixer** for both transmitter and receiver
- Students can use the MOSbius board to make a customizable mixer, emulate a natural working transmission channel*, and implement a bandpass filter to select which signal to process
- The chip is usable for transmitter and receiver
- We will get 1) **Phase-Locked Loop (PLL+VCO)** and 2) **Crystal Oscillator** using COTS components.
- We intend to implement the FSK modulator and demodulator on FPGA using existing designs and use a lab tool or an already-made **PCB** for the **AWGN Generator**

- **Application**

- Education and Experimentation: A hands-on platform for teaching students system-level principles of spread spectrum communication on a clean and real-life emulated channel. This chip uses the 51-58 MHz LO signal with 10.7 MHz FSK Modulation rate and +/-50 kHz frequency deviation and 100 kHz bit rate (fm).

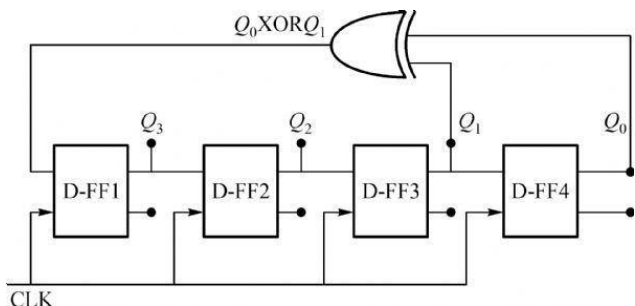
- **Early References**

- Example of DSSS/FHSS Trainer Kit: <https://www.tescaglobal.com/product/cdma-dsss-fhss-trainer>
- https://www.researchgate.net/publication/309521542_A_modified_closed-loop_auto_frequency_calibration_technique_for_a_15
- https://www.researchgate.net/publication/314818626_Design_and_Simulation_of_RADAR_Transmitter_and_Receiver_using_Direct_Sequence_Spread_Spectrum

PN Sequence Generator Implementation

We are going to make a **PN Sequence Generator** on the IC Layout Design. The PN Sequence Generator will generate a sequence of numbers that are going to corresponds to a specific frequency channel enabling the frequency hopping function of our system. Our MVP consists of a 4 flip-flop PN Generator (capable of accommodating 15 different channels using 10 kHz clock).

If we used the Si535X, we will likely implement it on-chip, but with an additional register to translate the PN_OUT (aka Q_{0-3}) to I2C Communication compatible with the Eval Board's driver.



4 Flip-Flop PN Sequence Generator

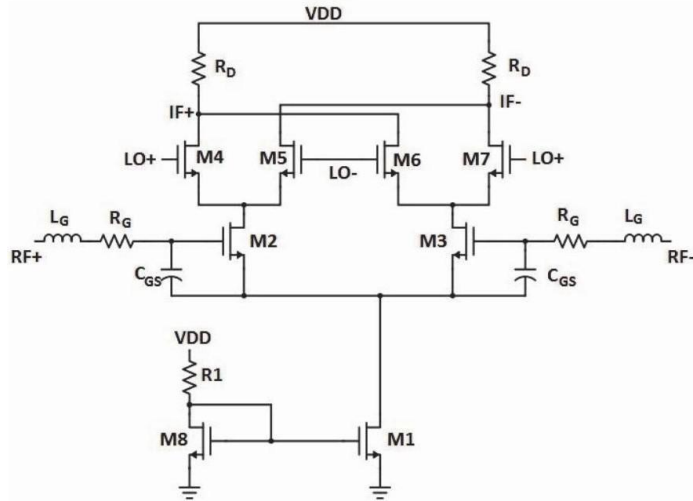
Pin	I/O	Purpose
CLK	Input	<u>System Clock</u> This is the "trigger." to make all the flip-flops of change state at the exact same time, keeping the system synchronized.
Q_{0-3}	Output	<u>Main Output</u> Output from the flip-flops forming the parallel channel number. (4-bit)
SEED ₀₋₃	Input	<u>Initializing Input</u> Initializing input so the pn sequencing can start (4-bit number except 0000)

Mixer Implementation

For the Mixer, we are going to use **Double-balanced Gilbert Cell**. The RF+ and RF- input will go through an **Impedance Matching Circuit** (R_G , L_G , C_{GS}) before going into the main Gilbert Cell. The specification of the Mixer will highly depends on the VCO+PLL output.

For the implementation, we will mainly use this paper as main reference:

https://www.researchgate.net/publication/358030720_Design_of_a_High-Linearity_Gilbert-Cell_Mixer_at_5_GHz



Double Balanced Gilbert Cell

Pin	I/O	Purpose
RF+, RF-	Input	<u>Radio Frequency</u> The travelling signal on then channel (at a low frequency).
LO+, LO-	Input	<u>Local Oscillator</u> The high-frequency hopping carrier from the VCO .
IF+, IF-	Output	<u>Intermediate Frequency</u> This will be the output for demodulation input (down conversion) or FSK modulated signal (upconversion). Ideally, it outputs a signal with a new frequency based on the difference of the RF frequency and LO frequency. $f_{IF} = f_{RF} - f_{LO} $

VCO and PLL Implementation (Si535X)

For the VCO and PLL, we're going to use the SI535X Chip via SI535X-B20QFN-EVB Eval Board. We choose the chip because it has features:

- Programmable Divider (Programmable using I2C)
- Integrated VCO
- Good frequency range (50-100 MHz)

The main problem with this chip is the divider programming for our frequency hopping is programmed using I2C interface. Also, the clock output is only one phase while the mixer needs two different phase signal input (180 deg difference)

<https://www.analog.com/media/en/technical-documentation/data-sheets/adf4351.pdf>

Alternative VCO and PLL Implementation

ADF4351 Chip

We have looked for alternative chips for our needs, and we found that ADF4351 also fits our criteria. However it came with a few caveats:

- Divider is programmable using SPI (Making PN Generator on-chip implementation rather difficult, easier to implement on FPGA instead)
- No eval board available that are on our working frequency. (We found one on 2.2 GHz - 4.4 GHz)

VCO and PLL IC Design

The first (original) plan was to implement the VCO in our IC Chip Design and connect it to PLL chip (COTS). We underestimated the complexity of implementing the VCO - PLL (a PLL chip that takes external VCO is hard to find and/or have a rather disappointing specifications).

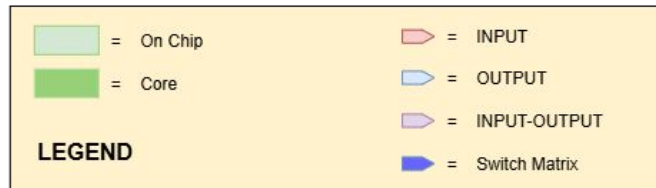
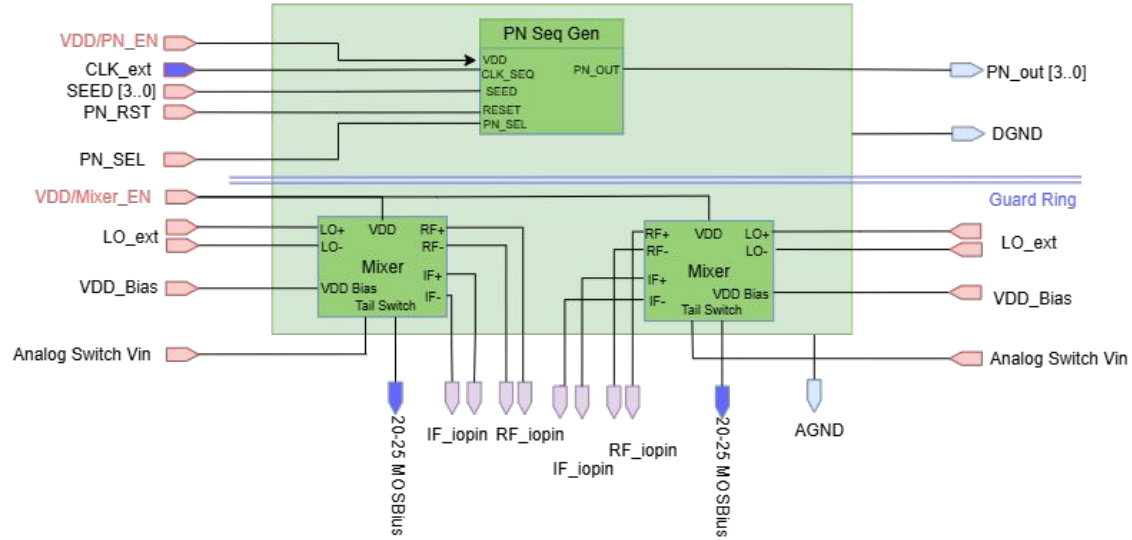
End Product

Target Product

- Input : FSK Modulated Signal
- Output : FSK Demodulated Signal
- Signal Frequency : **61-68 MHz**
- Hop Channel : **15 Channel**
- IC Blocks : **PN Sequence Generator, Gilbert Cell Mixer**
- MOSBius : **FHSS Transmitter -> AWGN Noise Generator -> Receiver**
- COTS Blocks : **VCO+PLL+Crystal Oscillator (Si535x Eval Board)**

Block Diagrams (Pin Out)

CHIP PINS



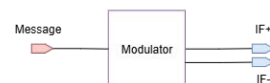
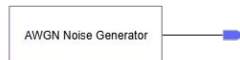
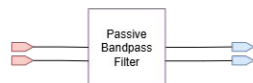
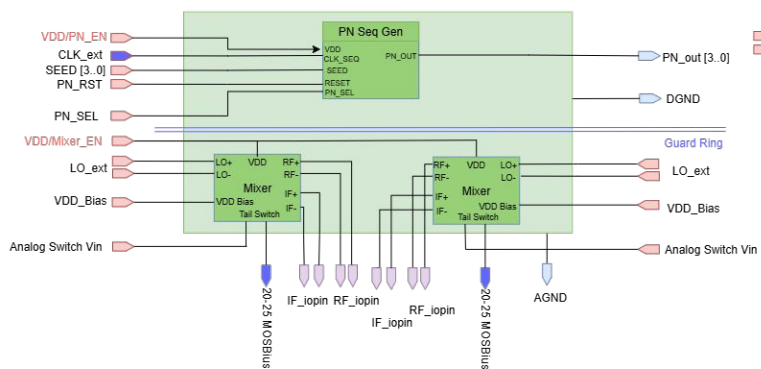
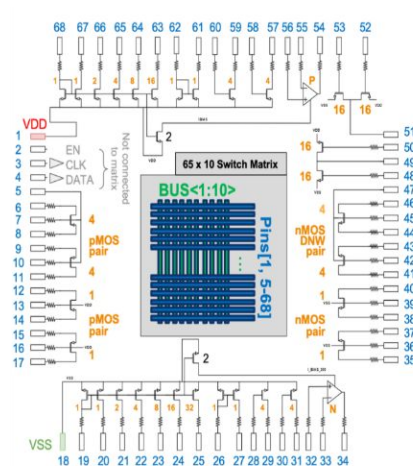
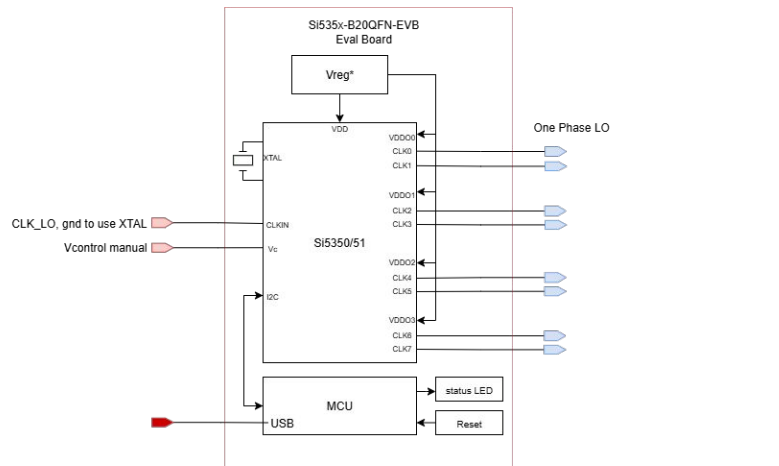
Block Diagrams (Pin List for Chip)

Pin #	Pin Name	Type	Description
Power & Ground			
1	VDD/Mixer_EN	Power	Positive power supply for the Mixer and Switch (Analog). Connect to load for gain modificaton.
2	VDD/PN_EN	Power	Positive power supply for the PN (digital).
3	DGND	Power	Ground reference for the PN (digital).
4	AGND	Power	Ground reference for the Mixer and Switch (analog).
RF, IF & LO Ports			
5-6	RF+	I/O	Non-inverting input & output for the Radio Frequency (RF) signal.
7-8	RF-	I/O	Inverting input & output for the Radio Frequency (RF) signal.
9-10	LO_ext+	I	Non-inverting input for the external Local Oscillator (LO) signal.
11-12	LO_ext-	I	Inverting Input for the external Local Oscillator (LO) signal.
13-14	IF+	I/O	Non-inverting input & output for the Intermediate Frequency (IF) signal.
15-16	IF-	I/O	Inverting input & output for the Intermediate Frequency (IF) signal.

Block Diagrams (Pin List for Chip)

Pin #	Pin Name	Type	Description
Bias, Control & Digital I/O			
17-18	VDD_Bias	I	Analog input. External bias voltage port. Allows user to tune the VDD_Bias to control the current source. Add resistance to modify gain.
19-20	Analog Switch Vin	I	Digital input. Controls the internal multiplexer to select the mixer's bias source. (LOW for internal source, HIGH for external source).
21-22	MOSBius Current Mirror	O	Analog output. Allows user to select mixer's tail current from MOSBius when selected by Analog Switch Vin.
23	CLK_ext/PN_EN	I	Digital input. External clock signal for the PN Sequence Generator block.
24-27	PN_out	O	Digital output. Provides the Pseudo-random Number (PN) sequence generated by the internal LUT/generator. 4 Bit.
28	PN_RST	I	Digital input. Reset PN Sequence to DFF
29	PN_SEL	I	Digital input. 2 MUX 1 for input XOR feedback. The XOR accepts Q of last DFF and one nearest or one furthest from the last DFF
30-33	Seed	I	Digital input. 4 Bit initializer.

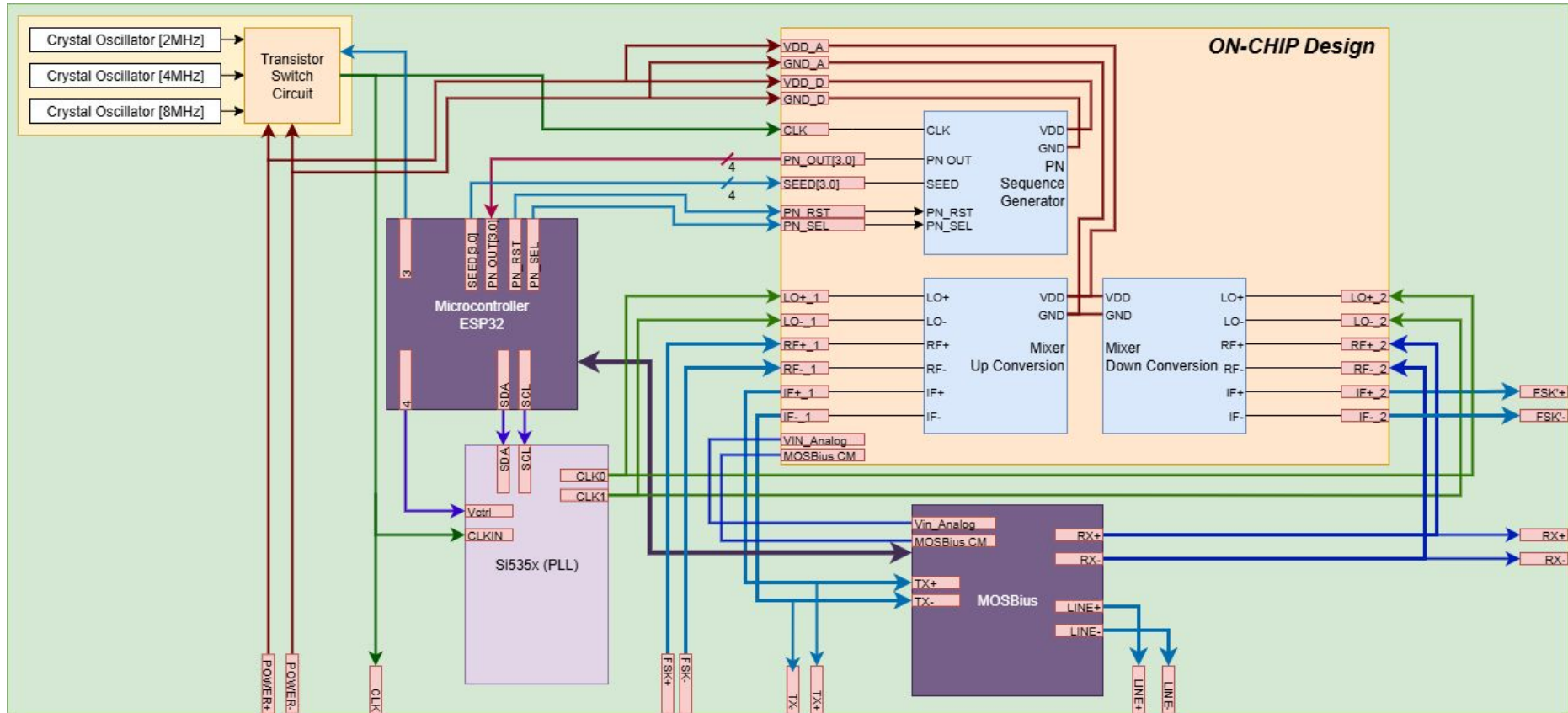
Block Diagrams (System level)



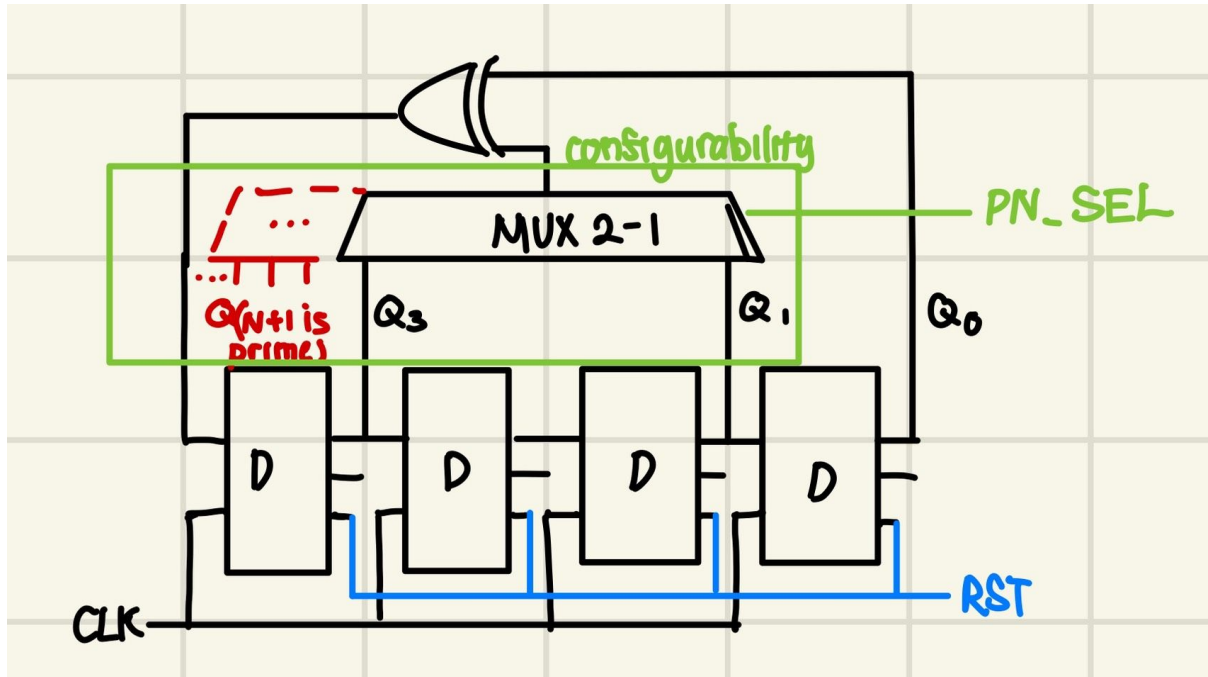
LEGEND

- On Chip
- Available/Provided Tool
- Built/Designed/Configured by our team
- COTS components
- INPUT
- OUTPUT
- INPUT-OUTPUT
- Switch Matrix

System Diagram



Configurable PN Sequence Generator

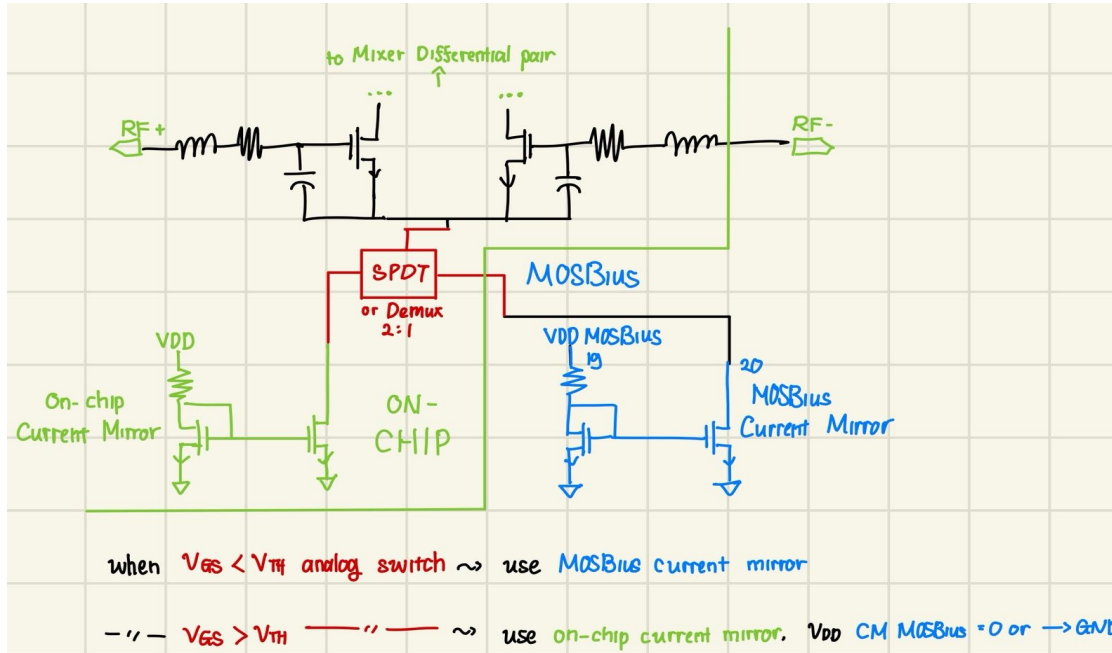


The XOR accepts Q of last DFF (output) and one Q_{N+1} where $N+1$ is prime number

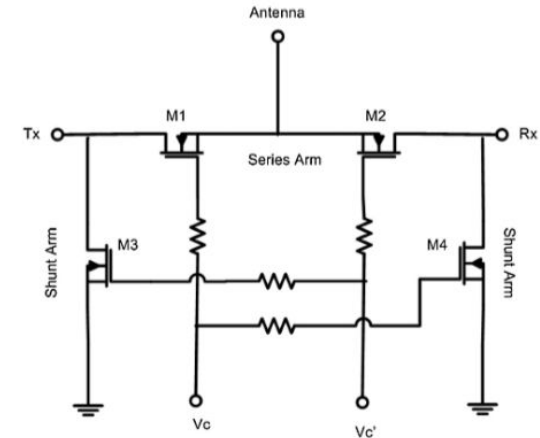
References:

1. https://www.researchgate.net/publication/371975741_Reconfigurable_linear_feedback_shift_register_for_wireless_communication_and_coding

Customizable Mixer (Current Mirror)



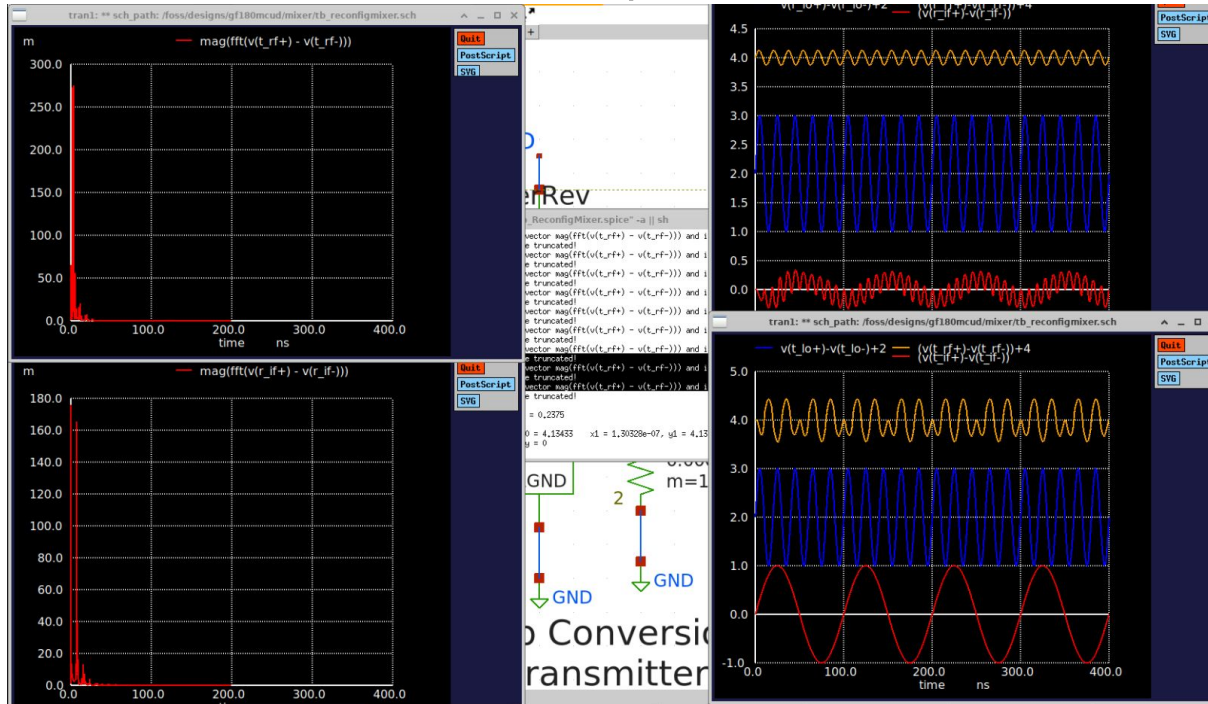
The analog switch will be implemented on-chip, mirroring the T/R switch concept (SPDT)



References:

1. <https://ieeexplore.ieee.org/document/4907310>
2. https://www.researchgate.net/publication/325621113_A_fully_integrated_high_IP1dB_CMOS_SPDT_switch_using_stacked_transistors_for_24_GHz_TD_D_transceiver_applications

Customizable Mixer (On-Chip Current Mirror)



Top: FFT and Tran for Downconversion

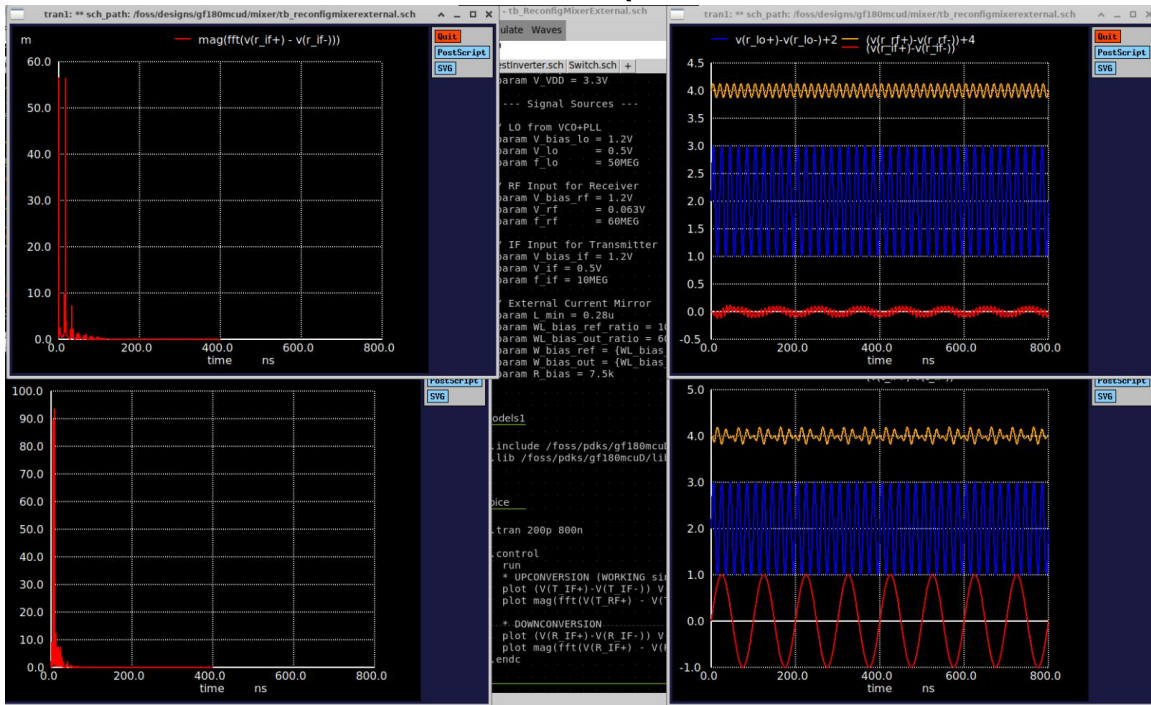
Bottom: FFT and Tran for UPconversion

Vin Switch = 3.3V

References:

1. <https://ieeexplore.ieee.org/document/4907310>
2. https://www.researchgate.net/publication/325621113_A_fully_integrated_high_IP1dB_CMOS_SPDT_switch_using_stacked_transistors_for_24_GHz_TD_D_transceiver_applications

Customizable Mixer (External Current Mirror)



Top: FFT and Tran for Downconversion

Bottom: FFT and Tran for UPconversion

Vin Switch = -3.3V

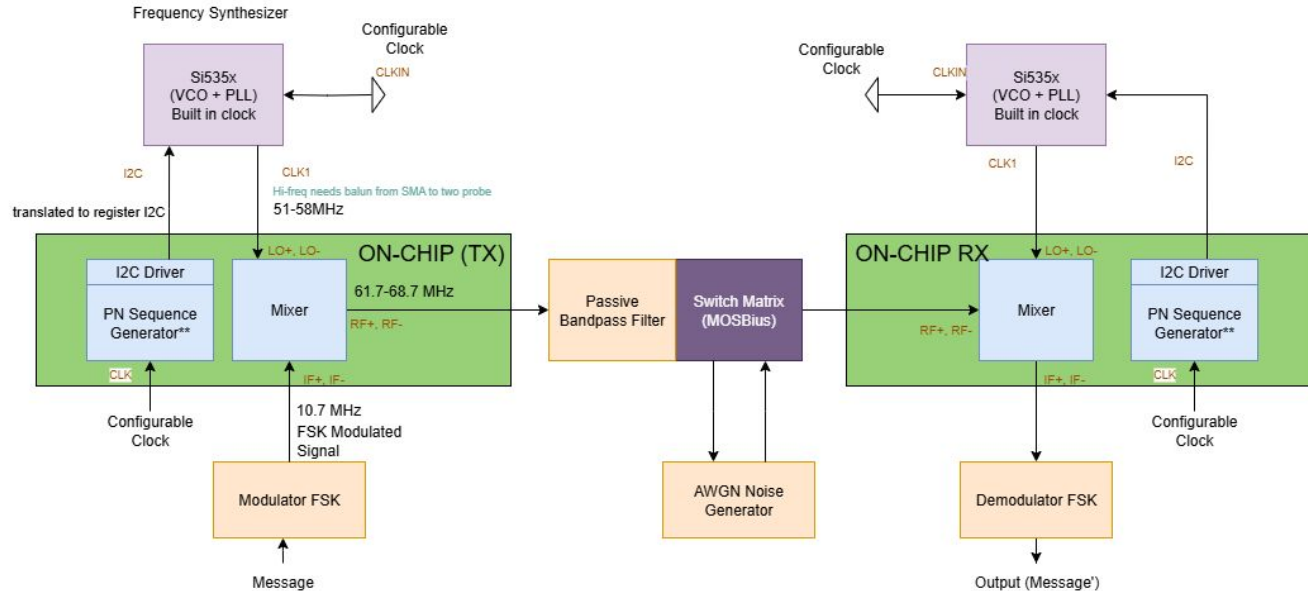
References:

1. <https://ieeexplore.ieee.org/document/4907310>
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Block Diagram Implementations (FHSS)

The FHSS Learning Kit System Architecture

The FHSS System



* Configurable Clocks are implemented using Analog Circuit with Circuit controlled using Switch Matrix or Physical Switch

* FSK Modulator and Demodulator are implemented using FPGA or MCU

Team Member Tasks

Tasks:

- Reconfigurable Mixer: James Patrick
- Reconfigurable PN: Ibrahim Hanif
- Switch Matrix Layout: Goldwin Sonick
- Top Level Integration + Proposal: All Members

If time allows (listed by sorting which to do first):

- Phase detector (repurpose the double balanced gilbert cell mixer)
- Modulator FSK
- BER computation (synchronised XOR from message to message')
- AWGN generator

Timeline

Week	Starting Date	Ending Date	Task	PIC	Expected Product	Details
Phase 1 : Setup and Introduction						
Phase 2 : Team Formation and Project Planning						
Phase 3 : Design and Simulation						
31	28/07/2025	03/08/2025	Xschem + Testbench Implementation	All members working in parallel	sch files, simulation	Schematic for: IC Block: PN Seq Generator, Mixer MOSBius : AWGN Noise Generator
32	04/08/2025	10/08/2025	Xschem + Testbench Implementation	All members working in parallel	sch files, simulation	
33	11/08/2025	17/08/2025	Layout Implementation	All members working in parallel	mag files, parasitic extraction	Layout for: IC Block: PN Seq Generator, Mixer

Timeline

Phase 4 : Layout and Verification						
34	18/08/2025	24/08/2025	Layout Implementation	All members working in parallel	mag files, parasitic extraction	Layout for: IC Block: PN Seq Generator, Mixer
35	25/08/2025	31/08/2025	Layout Implementation	All members working in parallel	mag files, parasitic extraction	
36	01/09/2025	07/09/2025	Top Level Integration	All members working in parallel	Top Level System	Integrating our system with MOSbius, designing the final pcb kit, checking interface
37	08/09/2025	14/09/2025	Top Level Integration	All members working in parallel	Top Level System	
38	15/09/2025	21/09/2025	Testing and Revisions	All members working in parallel	Top Level System	Final revisions
39	22/09/2025	28/09/2025	Testing and Revisions	All members working in parallel	Top Level System	Final Submission Day
40	29/09/2025	05/10/2025	Retrospection			Project Retrospective

Progress So Far

Week	Starting Date	Ending Date	Task	PIC	Checker	Details
Phase 1 : Setup and Introduction						
Phase 2 : Team Formation and Project Planning						
Phase 3 : Design and Simulation						
31	28/07/2025	03/08/2025	Per Block Schematic Design using Xschem + Testbench (MVP)	All members working in parallel	☒	Manually making each block using magic, extracting necessary files, then simulation using testbench for each block
			VCO	James	☒	Tuning range from 10-275MHz
			PN Generator	Ibrahim	☒	Successful Simulation, Improvement needed
			Mixer	Goldwin	☒	Responsibilities handover (switching jobdesk)
32	21/07/2025	27/07/2025	Per Block Schematic Design using Xschem + Testbench	All members working in parallel	☒	Making VCO, PN Sequence Generator and Mixer using Xschem, inc. simulation
			Mixer	James	☒	Modifying schematic to configurable design
			PN Generator	Ibrahim	☒	Successful Simulation, Improvement needed
			Mixer	Goldwin	☒	Responsibilities handover (switching jobdesk)
33	11/08/2025	17/08/2025	Top Level Verification, Design Optimization. + Layout start	All members working in parallel for top level verification	☐	Optimization and layout
			Mixer	James	☐	
			PN Generator	Ibrahim	☐	
			MOSbius circuits (switch matrix)	Goldwin	☐	
Phase 4 : Layout and Verification						
34	18/08/2025	24/08/2025	PCB, MOSbius, Interface Design + top level integration	All members working in parallel	☐	Integrating our system with MOSbius, designing the final pcb kit, checking interface
35	25/08/2025	31/08/2025	PCB, MOSbius, Interface Design + Top Level Verification, Design Optimization	All members working in parallel	☐	Integrating our system with MOSbius, designing the final pcb kit, checking interface
34	18/08/2025	24/08/2025	Layout Implementation	All members working in parallel	☐	mag files, parasitic extraction
35	25/08/2025	31/08/2025	Layout Implementation	All members working in parallel	☐	mag files, parasitic extraction
36	01/09/2025	07/09/2025	Top Level Integration	All members working in parallel	☐	Top Level System
37	08/09/2025	14/09/2025	Top Level Integration	All members working in parallel	☐	Top Level System
38	15/09/2025	21/09/2025	Testing and Revisions	All members working in parallel	☐	Top Level System
39	22/09/2025	28/09/2025	Testing and Revisions	All members working in parallel	☐	Top Level System
40	29/09/2025	05/10/2025	Retrospection			

Progress Tracker

Hierarchy level	Cell	Schematic-level design+sims	Layout	Post-layout sims
1	PN Sequence Generator	80.00%	0.00%	0.00%
1	Gilbert Cell Mixer	90.00%	0.00%	0.00%
2	Configurable XOR feedback	90.00%	0.00%	0.00%
2	Configurable SEED	20.00%	0.00%	0.00%
2	Selectable Current Mirror	20.00%	0.00%	0.00%
3	chip_top (On-Chip Integration)	0.00%	0.00%	0.00%
4	FHSS system verification	0.00%	0.00%	0.00%
System	System Integration (PCB, COTS, MOSBius)	0.00%	0.00%	0.00%